Decision-Making Under Uncertainty: Exam Solutions

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Learning objective

Structure decision-making problems under uncertainty, calculate and organise payoffs, and apply the Maximax and Maximin criteria to compare alternatives and justify decisions.

Criteria assessment

This task sheet assesses criteria C1 through C4. Each problem assesses the criteria indicated in its title. The number of points assigned to each task is shown at the end of the item. Partial credit may be awarded when the correct procedure is shown but arithmetic errors are made. Answers that do not show the required procedure will not receive credit. A minimum of 4 points is required to pass each criterion.

Evaluated criteria

C1 Characterises decision-making problems under uncertainty by identifying decision alternatives, states of nature, consequences, and the economic or financial context of the decision.

C2 Organises decision problems using payoff tables, placing alternatives, states of nature, and payoffs in a clear structure for comparison.

C3 Calculates profits by combining revenues with fixed and variable costs across different states of nature, determining the corresponding payoffs for each alternative, and organising the results in payoff tables for comparison.

C4 Applies the Maximin and Maximax criteria to make optimistic or conservative decisions without probabilities, selecting the alternative with the highest possible payoff or the best worst-case, interpreting risks or limitations.

1 Local Food Market Stall [C1-1 C2-2]

A local food market company is planning its stall offering for an upcoming market period. A standard stall is expected to generate profits of eighteen thousand US dollars under high demand and eleven thousand US dollars under low demand, while a deluxe stall is expected to generate thirty-one thousand US dollars and six thousand US dollars, respectively. The company must select its stall before the actual level of customer demand is known.

Questions/Tasks.

C1 Characterize the key elements of the decision-making situation. **[1]**
C2 Construct the payoff table. **[2]**

C1 - Characterization [1]

Element	Description	
Alternatives	**A1:** Standard stall	**A2:** Deluxe stall
Statesofnature	**S1:** High demand	**S2:** Low demand
Events	Actual market demand after the stall decision	
Consequences	Profit in thousand US dollars for each alternative–state pair	

C2 - Payoff table [2]

Each stated consequence is placed at the intersection of its alternative and state:

$$P(A_1, S_1) = 18, \quad P(A_1, S_2) = 11,$$
$$P(A_2, S_1) = 31, \quad P(A_2, S_2) = 6.$$

Thus, in thousand US dollars, the completed payoff table is

Alternative	High S_1	Low S_2
Standard stall A_1	18	11
Deluxe stall A_2	31	6



2 Small Bakery Production [C1-1 C2-3]

A small bakery is planning its production capacity for an upcoming sales period. A standard oven can generate 3 thousand BRL per production batch, while an industrial oven can generate 5 thousand BRL per batch. Based on expected demand, the standard oven would complete 16 batches under high demand or 11 under low demand, whereas the industrial oven

would complete 13 or 9 batches, respectively. In addition, the bakery would receive a fixed sales bonus of 6 thousand BRL if demand is high and incur a fixed loss of 3 thousand BRL if demand is low.

Questions/Tasks.

C1 Characterize the key elements of the decision-making situation. **[1]**

C2 Construct the payoff table. **[3]**

C1 - Characterization [1]

Element	Description	
Alternatives	**A1:** Standard oven	**A2:** Industrial oven
Statesofnature	**S1:** High demand	**S2:** Low demand
Events	Actual customer demand after the oven decision	
Consequences	Profit in thousand BRL for each alternative–state pair	

C2 - Payoff table [3]

Relevant values.

Abstract component	Context component	Math symbol	Corresponding value
Quantity, alternative–state dependent	Expected completed production batches for each oven under each demand condition	$q(A_i, S_j)$	Standard–high: 16 batches; Standard–low: 11 batches; Industrial–high: 13 batches; Industrial–low: 9 batches
Variable, alternative dependent	Earnings per completed production batch from the chosen oven	$v^A(A_i)$	Standard oven: 3 thousand BRL/batch; Industrial oven: 5 thousand BRL/batch
Variable, state dependent	Additional contribution per completed production batch from the demand condition	$v^S(S_j)$	High demand: 0 thousand BRL/batch; Low demand: 0 thousand BRL/batch
Fixed, state dependent	Fixed sales bonus or loss associated with the demand condition	$f^S(S_j)$	High demand: 6 thousand BRL; Low demand: -3 thousand BRL

Payoff equation.

$$P_{ay}(A_i, S_j) = q(A_i, S_j)v^A(A_i) + f^S(S_j).$$

Calculations (thousand BRL).

Oven	High demand	Low demand
Standard oven	$16(3) + 6 = 54$	$11(3) - 3 = 30$
Industrial oven	$13(5) + 6 = 71$	$9(5) - 3 = 42$

Final payoff table (profit in thousand BRL).

Oven	High demand	Low demand
Standard oven	54	30
Industrial oven	71	42



3 Urban Meal Subscription Service [C1-2 C2-5]

An urban meal subscription company must choose between a basic meal plan and a premium meal plan before customer interest is known. Interest may be strong or weak. The basic plan generates 4 thousand BRL per active subscriber and also provides a fixed operating contribution of 6 thousand BRL. The premium plan generates 6 thousand BRL per active subscriber and provides a fixed operating contribution of 9 thousand BRL. Strong interest adds another 2 thousand BRL per active subscriber to either plan, while weak interest adds 1 thousand BRL per active subscriber. The basic plan is expected to have 13 active subscribers under strong interest and 8 under weak interest; the premium plan is expected to have 10 active subscribers under strong interest and 6 under weak interest.

Questions/Tasks.

C1 Characterize the key elements of the decision-making situation. **[2]**
C2 Construct the payoff table. **[5]**

C1 - Characterization [2]

Element	Description	
Alternatives	**A1:** Basic meal plan	**A2:** Premium meal plan
Statesofnature	**S1:** Strong customer interest	**S2:** Weak customer interest
Events	Actual customer interest after the meal plan is selected	
Consequences	Profit in thousand BRL for each alternative–state pair	

C2 - Payoff table [5]

Relevant values.

Abstract component	Context component	Math symbol	Corresponding value
Quantity, alternative–state dependent	Expected active subscribers for each meal plan under each level of customer interest	$q(A_i, S_j)$	Basic–strong: 13 subscribers; Basic–weak: 8 subscribers; Premium–strong: 10 subscribers; Premium–weak: 6 subscribers
Variable, alternative dependent	Contribution per active subscriber from the chosen meal plan	$v^A(A_i)$	Basic plan: 4 thousand BRL/subscriber; Premium plan: 6 thousand BRL/subscriber
Variable, state dependent	Additional contribution per active subscriber from customer interest	$v^S(S_j)$	Strong interest: 2 thousand BRL/subscriber; Weak interest: 1 thousand BRL/subscriber
Fixed, alternative dependent	Fixed operating contribution from the chosen meal plan	$f^A(A_i)$	Basic plan: 6 thousand BRL; Premium plan: 9 thousand BRL

Payoff equation.

$$P_{ay}(A_i, S_j) = q(A_i, S_j)\left[v^A(A_i) + v^S(S_j)\right] + f^A(A_i).$$

Calculations (thousand BRL).

Meal plan	Strong customer interest	Weak customer interest
Basic plan	$13(4+2)+6=84$	$8(4+1)+6=46$
Premium plan	$10(6+2)+9=89$	$6(6+1)+9=51$

Final payoff table (profit in thousand BRL).

Meal plan	Strong customer interest	Weak customer interest
Basic plan	84	46
Premium plan	89	51



4 Regional Bookstore Expansion [C1-1 C3-2]

A regional bookstore must decide whether to open a children's section or maintain its current product selection before customer traffic is known. Traffic may be strong or weak. All values are in million BRL. Opening the children's section produces revenue of 72 under strong traffic and 35 under weak traffic, while maintaining the current selection produces revenue of 61 under strong traffic and 47 under weak traffic. The corresponding costs are 24 and 22 for the children's section under strong and weak traffic, respectively, and 25 and 23 for the current selection under strong and weak traffic, respectively.

Questions/Tasks.

C1 Characterize the key elements of the decision-making situation. **[1]**

C3 Use revenue and costs to construct the payoff table. **[2]**

C1 - Characterization [1]

Element	Description	
Alternatives	**A1:** Open children's section	**A2:** Maintain current selection
Statesofnature	**S1:** Strong traffic	**S2:** Weak traffic
Events	Actual customer traffic after the product selection decision	
Consequences	Profit in million BRL for each alternative–state pair	

C3 - Calculating profit and building the payoff table [2]

Relevant values.

Abstract component	Context component	Math symbol	Corresponding value
Revenue, alternative–state dependent	Total revenue generated by each product selection under each traffic condition	$R(A_i, S_j)$	Children's section–strong: 72 million BRL; Children's section–weak: 35 million BRL; Current selection–strong: 61 million BRL; Current selection–weak: 47 million BRL
Cost, alternative–state dependent	Total cost associated with each product selection under each traffic condition	$C(A_i, S_j)$	Children's section–strong: 24 million BRL; Children's section–weak: 22 million BRL; Current selection–strong: 25 million BRL; Current selection–weak: 23 million BRL

Profit equation.

$$P_{ay}(A_i, S_j) = R(A_i, S_j) - C(A_i, S_j).$$

Calculations (million BRL).

Alternative and state	Revenue	Cost	Profit
Children's section–strong traffic	72	24	$72 - 24 = 48$
Children's section–weak traffic	35	22	$35 - 22 = 13$
Current selection–strong traffic	61	25	$61 - 25 = 36$
Current selection–weak traffic	47	23	$47 - 23 = 24$

Final payoff table (profit in million BRL).

Alternative	Strong traffic	Weak traffic
Open children's section	48	13
Maintain current selection	36	24

5 Greenhouse Crop Production [C1-1 C3-3]

A greenhouse farm is planning its production for an upcoming growing season. Revenue per production lot is 18 thousand EUR for tomatoes and 15 thousand EUR for peppers. Under favourable weather conditions, the farm expects to produce 8 lots of tomatoes and 7 lots of peppers, while under unfavourable weather conditions it expects to produce 5 lots of tomatoes and 4 lots of peppers. The cost per production lot is 7 thousand EUR under favourable weather and 10 thousand EUR under unfavourable weather.

Questions/Tasks.

C1 Characterize the key elements of the decision-making situation. **[1]**
C3 Use revenue and costs to construct the payoff table. **[3]**

C1 - Characterization [1]

Element	Description	
Alternatives	**A1:** Produce tomatoes	**A2:** Produce peppers
Statesofnature	**S1:** Favourable weather	**S2:** Unfavourable weather
Events	Weather conditions revealed after the production decision	
Consequences	Profit in thousand EUR for each crop–weather-state pair	

C3 - Profit and payoff table [3]

Relevant values.

Abstract component	Context component	Math symbol	Corresponding value
Quantity, alternative–state dependent	Production lots for each crop under each weather condition	$q(A_i, S_j)$	Tomatoes–favourable: 8; Tomatoes–unfavourable: 5; Peppers–favourable: 7; Peppers–unfavourable: 4
Variable, alternative dependent	Revenue per production lot	$r(A_i)$	Tomatoes: 18 thousand EUR/lot; Peppers: 15 thousand EUR/lot
Variable, state dependent	Cost per production lot	$c(S_j)$	Favourable weather: 7 thousand EUR/lot; Unfavourable weather: 10 thousand EUR/lot

Profit equation.

$$R = q(A_i, S_j)r(A_i), \qquad C = q(A_i, S_j)c(S_j), \qquad P_{ay}(A_i, S_j) = R - C.$$

Calculations (thousand EUR).

Cell	Revenue	Cost	Profit
(A_1, S_1)	$8(18) = 144$	$8(7) = 56$	$144 - 56 = 88$
(A_1, S_2)	$5(18) = 90$	$5(10) = 50$	$90 - 50 = 40$
(A_2, S_1)	$7(15) = 105$	$7(7) = 49$	$105 - 49 = 56$
(A_2, S_2)	$4(15) = 60$	$4(10) = 40$	$60 - 40 = 20$

Final payoff table (profit in thousand EUR).

Alternative	Favourable weather	Unfavourable weather
Produce tomatoes	88	40
Produce peppers	56	20

6 Fitness Center Membership Plans [C1-2 C3-5]

A fitness center must choose between a basic membership plan and a premium membership plan before monthly demand is known. Demand may be high or low. Quantities, in hundreds of memberships, are 9 and 6 for the basic plan and 7 and 4 for the premium plan under high and low demand, respectively. The plan-based revenue rates are 8 and 13 thousand USD per hundred memberships for the basic and premium plans. High demand adds a state-based revenue rate of 2 thousand USD per hundred memberships; low demand adds 1. The plan-based cost rates are 3 and 6 thousand USD per hundred memberships, respectively. The fixed operating costs are 12 thousand USD under high demand and 16 thousand USD under low demand.

Questions/Tasks.

C1 Characterize the key elements of the decision-making situation. **[2]**
C3 Use revenue and costs to construct the payoff table. **[5]**

C1 - Characterization [2]

Element	Description	
Alternatives	**A1:** Offer a basic membership plan	**A2:** Offer a premium membership plan
Statesofnature	**S1:** High monthly demand	**S2:** Low monthly demand
Events	Monthly market conditions revealed after the membership-plan decision	
Consequences	Profit in thousand USD for each plan–market-state pair	

C3 - Profit and payoff table [5]



Relevant values.

Abstract component	Context component	Math symbol	Corresponding value
Quantity, alternative–state dependent	Membership lots for each plan under each demand level	$q(A_i, S_j)$	Basic–high: 9; Basic–low: 6; Premium–high: 7; Premium–low: 4 hundreds of memberships
Variable, alternative dependent	Plan-based revenue rate per hundred memberships	$r^A(A_i)$	Basic plan: 8 thousand USD/100 memberships; Premium plan: 13 thousand USD/100 memberships
Variable, state dependent	Additional revenue rate per hundred memberships from monthly demand	$r^S(S_j)$	High demand: 2 thousand USD/100 memberships; Low demand: 1 thousand USD/100 memberships
Variable, alternative dependent	Plan-based cost rate per hundred memberships	$c^A(A_i)$	Basic plan: 3 thousand USD/100 memberships; Premium plan: 6 thousand USD/100 memberships
Fixed, state dependent	Fixed operating cost under each demand condition	$f_c(S_j)$	High demand: 12 thousand USD; Low demand: 16 thousand USD

Profit equation.

$$R = q(A_i, S_j)\left[r^A(A_i) + r^S(S_j)\right], \qquad C = q(A_i, S_j)c^A(A_i) + f_c(S_j), \qquad P_{ay}(A_i, S_j) = R - C.$$

Calculations (thousand USD).

Cell	Revenue	Cost	Profit
(A_1, S_1)	$9(8+2) = 90$	$9(3) + 12 = 39$	$90 - 39 = 51$
(A_1, S_2)	$6(8+1) = 54$	$6(3) + 16 = 34$	$54 - 34 = 20$
(A_2, S_1)	$7(13+2) = 105$	$7(6) + 12 = 54$	$105 - 54 = 51$
(A_2, S_2)	$4(13+1) = 56$	$4(6) + 16 = 40$	$56 - 40 = 16$

Final payoff table (profit in thousand USD).

Alternative	High monthly demand	Low monthly demand
Basic membership plan	51	20
Premium membership plan	51	16

7 Restaurant Expansion Strategy [C4-2]

A restaurant company must choose a strategy for expanding into a new location. Customer demand could be high, medium, or low. All payoffs are in thousands of dollars.

	Demand		
Alternative	High	Medium	Low
Full-service restaurant	78	35	8
Casual restaurant	72	48	32
Express restaurant	95	42	28
Postpone expansion	25	25	25

Questions/Tasks.

C4 What would the optimistic decision be (Maximax)? What would the conservative decision be (Maximin)? **[2]**

C4 - Maximax and Maximin **[2]**



Complete the table using letters first.

	A_1	A_2	A_3	A_4	Decision
Max	M_1	M_2	M_3	M_4	$A^*_{\text{Maximax}} = A_3$
Min	m_1	m_2	m_3	m_4	$A^*_{\text{Maximin}} = A_2$

Replace the letters with their values.

	A_1	A_2	A_3	A_4	Decision
Max	78	72	95	25	$A^*_{\text{Maximax}} = A_3$
Min	8	32	28	25	$A^*_{\text{Maximin}} = A_2$

Maximax

$$\begin{aligned} M_1 &= \max\{78, 35, 8\} = 78 \\ M_2 &= \max\{72, 48, 32\} = 72 \\ M_3 &= \max\{95, 42, 28\} = 95 \\ M_4 &= \max\{25, 25, 25\} = 25 \end{aligned}$$

Maximin

$$\begin{aligned} m_1 &= \min\{78, 35, 8\} = 8 \\ m_2 &= \min\{72, 48, 32\} = 32 \\ m_3 &= \min\{95, 42, 28\} = 28 \\ m_4 &= \min\{25, 25, 25\} = 25 \end{aligned}$$

Therefore, the optimistic decision is

$$A^*_{\text{Maximax}} = A_3 \quad \text{(Express restaurant)}$$

and the conservative decision is

$$A^*_{\text{Maximin}} = A_2 \quad \text{(Casual restaurant).}$$

8 Hotel Development Strategy **[C4-2]**

A hotel company must choose a strategy for developing a new property. Tourist demand could be high, medium, or low. All payoffs are in thousands of dollars.

	Tourist demand		
Alternative	High	Medium	Low
Large hotel	72	34	10
Medium hotel	65	46	22
Small hotel	95	44	31
No development	58	40	35

Questions/Tasks.

C4 What would the optimistic decision be (Maximax)? What would the conservative decision be (Maximin)? **[2]**

C4 - Maximax and Maximin **[2]**

Complete the table using letters first.

	A_1	A_2	A_3	A_4	Decision
Max	M_1	M_2	M_3	M_4	$A^*_{\text{Maximax}} = A_3$
Min	m_1	m_2	m_3	m_4	$A^*_{\text{Maximin}} = A_4$

Maximax

$$\begin{aligned} M_1 &= \max\{72, 34, 10\} = 72 \\ M_2 &= \max\{65, 46, 22\} = 65 \\ M_3 &= \max\{95, 44, 31\} = 95 \\ M_4 &= \max\{58, 40, 35\} = 58 \end{aligned}$$

Maximin

$$\begin{aligned} m_1 &= \min\{72, 34, 10\} = 10 \\ m_2 &= \min\{65, 46, 22\} = 22 \\ m_3 &= \min\{95, 44, 31\} = 31 \\ m_4 &= \min\{58, 40, 35\} = 35 \end{aligned}$$

Replace the letters with their values.

	A_1	A_2	A_3	A_4	Decision
Max	72	65	95	58	$A^*_{\text{Maximax}} = A_3$
Min	10	22	31	35	$A^*_{\text{Maximin}} = A_4$

Therefore, the optimistic decision is

$$A^*_{\text{Maximax}} = A_3 \quad \text{(Small hotel)}$$

and the conservative decision is

$$A^*_{\text{Maximin}} = A_4 \quad \text{(No development)}.$$

9 Agricultural Technology Investment Strategy **[C4-2]**

An agricultural company must choose a strategy for introducing a new automated farming system. Market conditions could be highly favourable, favourable, stable, or unfavourable. All payoffs are in thousands of dollars.

	Market conditions			
Alternative	Highly favourable	Favourable	Stable	Unfavourable
Full automation	125	58	8	-42
Partial automation	88	64	30	4
Basic automation	62	52	38	20
Equipment leasing	48	43	35	28

Questions/Tasks.

C4 What would the optimistic decision be (Maximax)? What would the conservative decision be (Maximin)? **[2]**

C4 - Maximax and Maximin **[2]**



Complete the table using letters first.

	A_1	A_2	A_3	A_4	Decision
Max	M_1	M_2	M_3	M_4	$A^*_{\text{Maximax}} = A_1$
Min	m_1	m_2	m_3	m_4	$A^*_{\text{Maximin}} = A_4$

Maximax

$$\begin{aligned} M_1 &= \max\{125, 58, 8, -42\} = 125 \\ M_2 &= \max\{88, 64, 30, 4\} = 88 \\ M_3 &= \max\{62, 52, 38, 20\} = 62 \\ M_4 &= \max\{48, 43, 35, 28\} = 48 \end{aligned}$$

Maximin

$$\begin{aligned} m_1 &= \min\{125, 58, 8, -42\} = -42 \\ m_2 &= \min\{88, 64, 30, 4\} = 4 \\ m_3 &= \min\{62, 52, 38, 20\} = 20 \\ m_4 &= \min\{48, 43, 35, 28\} = 28 \end{aligned}$$

Replace the letters with their values.

	A_1	A_2	A_3	A_4	Decision
Max	125	88	62	48	$A^*_{\text{Maximax}} = A_1$
Min	-42	4	20	28	$A^*_{\text{Maximin}} = A_4$

Therefore, the optimistic decision is

$$A^*_{\text{Maximax}} = A_1 \quad \text{(Full automation)}$$

and the conservative decision is

$$A^*_{\text{Maximin}} = A_4 \quad \text{(Equipment leasing).}$$

10 Summarized Solutions [C1-C4]

10.1 Local Food Market Stall C1–1 C2–2

Element	Description
Alt.	**A1:** Standard stall **A2:** Deluxe stall
States	**S1:** High demand **S2:** Low demand
Events	Actual market demand after the stall decision
Cons.	Profit in thousand US dollars for each alternative–state pair

Alt.	$\mathbf{S_1}$	$\mathbf{S_2}$
$\mathbf{A_1}$	18	11
$\mathbf{A_2}$	31	6

10.2 Small Bakery Production C1–1 C2–3

Element	Description
Alt.	**A1:** Standard oven **A2:** Industrial oven
States	**S1:** High demand **S2:** Low demand
Events	Actual customer demand after the oven decision
Cons.	Profit in thousand BRL for each alternative–state pair

Alt.	$\mathbf{S_1}$	$\mathbf{S_2}$
$\mathbf{A_1}$	$16(3)+6 = 54$	$11(3)-3 = 30$
$\mathbf{A_2}$	$13(5)+6 = 71$	$9(5)-3 = 42$

10.3 Urban Meal Subscription Service C1–2 C2–5

Element	Description
Alt.	**A1:** Basic meal plan **A2:** Premium meal plan
States	**S1:** Strong customer interest **S2:** Weak customer interest
Events	Actual customer interest after the meal plan is selected
Cons.	Profit in thousand BRL for each alternative–state pair

Alt.	$\mathbf{S_1}$	$\mathbf{S_2}$
$\mathbf{A_1}$	$13(4+2)+6 = 84$	$8(4+1)+6 = 46$
$\mathbf{A_2}$	$10(6+2)+9 = 89$	$6(6+1)+9 = 51$

10.4 Regional Bookstore Expansion C1–1 C3–2

Element	Description
Alt.	**A1:** Open children's section **A2:** Maintain current selection
States	**S1:** Strong traffic **S2:** Weak traffic
Events	Actual customer traffic after the product selection decision
Cons.	Profit in million BRL for each alternative–state pair

Alt.	$\mathbf{S_1}$	$\mathbf{S_2}$
$\mathbf{A_1}$	$72-24 = 48$	$35-22 = 13$
$\mathbf{A_2}$	$61-25 = 36$	$47-23 = 24$

10.5 Greenhouse Crop Production C1–1 C3–3

Element	Description
Alt.	**A1:** Produce tomatoes **A2:** Produce peppers
States	**S1:** Favourable weather **S2:** Unfavourable weather
Events	Weather conditions revealed after the production decision
Cons.	Profit in thousand EUR for each crop–weather-state pair

Alt.	$\mathbf{S_1}$	$\mathbf{S_2}$
$\mathbf{A_1}$	$8(18) - 8(7) = 88$	$5(18) - 5(10) = 40$
$\mathbf{A_2}$	$7(15) - 7(7) = 56$	$4(15) - 4(10) = 20$

10.6 Fitness Center Membership Plans C1–2 C3–5

Element	Description
Alt.	**A1:** Offer a basic membership plan **A2:** Offer a premium membership plan
States	**S1:** High monthly demand **S2:** Low monthly demand
Events	Monthly market conditions revealed after the membership-plan decision
Cons.	Profit in thousand USD for each plan–market-state pair

Alt.	$\mathbf{S_1}$	$\mathbf{S_2}$
$\mathbf{A_1}$	$9(8+2) - [9(3) + 12] = 51$	$6(8+1) - [6(3) + 16] = 20$
$\mathbf{A_2}$	$7(13+2) - [7(6) + 12] = 51$	$4(13+1) - [4(6) + 16] = 16$

10.7 Restaurant Expansion Strategy C4–2

	Max	Min
A_1	78	8
A_2	72	32
A_3	95	28
A_4	25	25
A^*_{Maximax}	A_3	
A^*_{Maximin}		A_2

10.8 Hotel Development Strategy C4–2

	Max	Min
A_1	72	10
A_2	65	22
A_3	95	31
A_4	58	35
A^*_{Maximax}	A_3	
A^*_{Maximin}		A_4

10.9 Agricultural Technology Investment Strategy C4–2

	Max	Min
A_1	125	-42
A_2	88	4
A_3	62	20
A_4	48	28
A^*_{Maximax}	A_1	
A^*_{Maximin}		A_4